

Retail: Customer Purchase Behavior & Sales Performance



Student Name:

Haneen Khalid

Nahla Hany

Abdulrahman Taher

Habiba Hamdy

Ahmed Yasser

Team

Name:G1

Introduction

The online retailer wants to better understand its sales performance and customer purchasing behavior.

The company needs to identify top-performing products, sales trends, high-value customers, and the countries generating the most revenue.

Our analysis provides data-driven insights to support better marketing and inventory planning for the next quarter.





Problem Statement

The challenge: The retailer has a massive volume of sales and customer data but lacks actionable insights

The Need: Better understanding of sales trends, top products ,and buyer .patterns

The Goal: Make data-driven marketing and inventory decisions for the upcoming quarter.



Problem Impact

The lack of proper analysis can lead to:

- Poor inventory planning
- Inefficient marketing decisions
- Missed sales opportunities
- Difficulty identifying valuable customers
- Reduced business performance



Project Objectives

- * Understand sales trends and overall business performance.**
- * Identify the top-performing products by revenue and quantity sold.**
- * Analyze customer purchasing behavior and order patterns.**
- * Identify high-value, at-risk, and new customers using RFM analysis.**
- * Identify key markets and seasonal patterns to support marketing and inventory planning.**



Exploration

What We Explored

- **Dataset structure and size**
- **Transactions, customers, products, and countries**
- **Date range covered by the dataset**
- **Date rangeaset covered by the dat**
- **Revenue distribution and overall sales patterns**
- **Sales performance across countries**
- **Product and customer contribution to revenue**
- **Revenue trends over time**



Data Cleaning

What We Cleaned

- **Standardized column names and formatting**
- **Removed unnecessary fields and irrelevant data**
- **Standardized data formats and values**
- **Removed duplicate records**
- **Separated cancelled transactions for separate analysis**
- **Removed transactions with invalid prices (≤ 0)**
- **Removed transactions with invalid quantities (≤ 0)**
- **Standardized product descriptions using XLOOKUP**

Which countries generate the highest revenue?

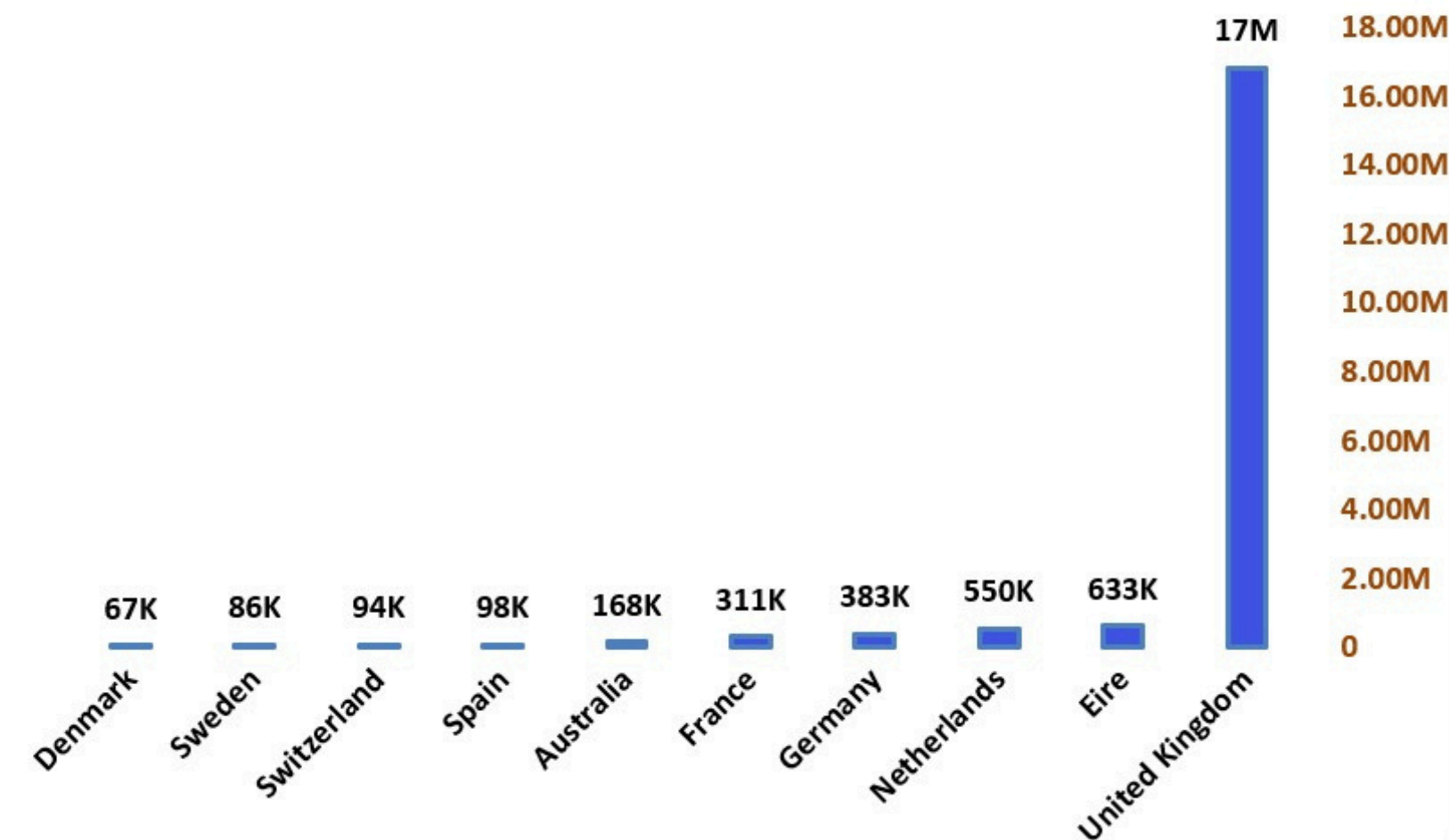
Insight

The UK generates approximately 87% of total revenue, showing a strong dependency on the British market and creating concentration risk.

Solution

Maintain the strong UK market while developing Ireland and the Netherlands as alternative growth markets to diversify revenue and reduce dependency on the UK.

Revenue by Country



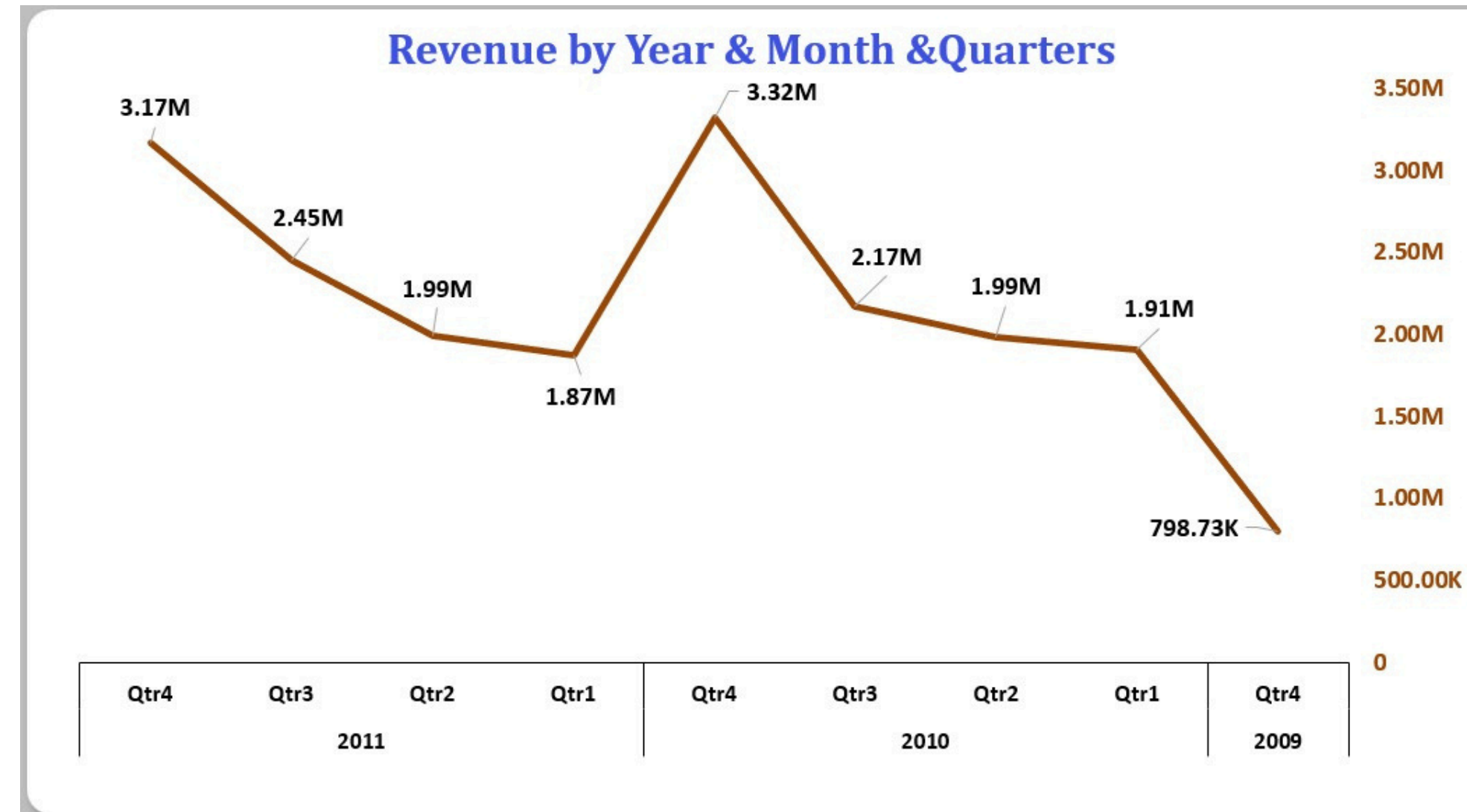
How does the company's revenue perform across the two years?

Insight

Annual revenue remained almost stable, with 9.40M in 2010 compared to 9.49M in 2011, indicating limited year-over-year .growth

Solution

Increase average customer spending through upselling, cross-selling, and product bundles.



How do monthly revenues trend across the two years?

Insight

Revenue peaks significantly in October and November. Also, Paper Chain Kit 50's Christmas is among the top 10 revenue-generating products, indicating clear (seasonality) in customer demand.

Solution

Increase stock levels of seasonal/Christmas products before October to meet peak demand and avoid stockouts.

Insight

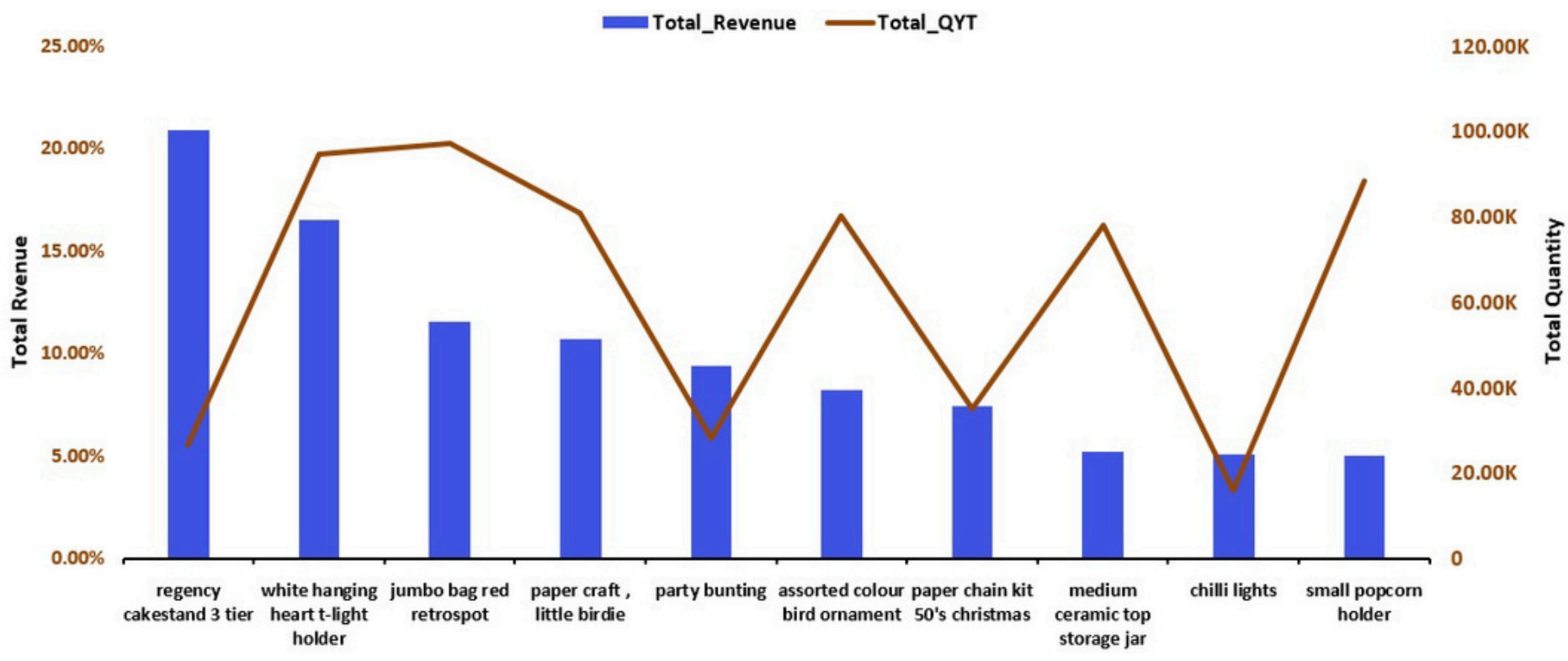
February was one of the weakest revenue months in both years: 538K in 2010 vs. 508K in 2011, showing a recurring low-demand period.

Solution

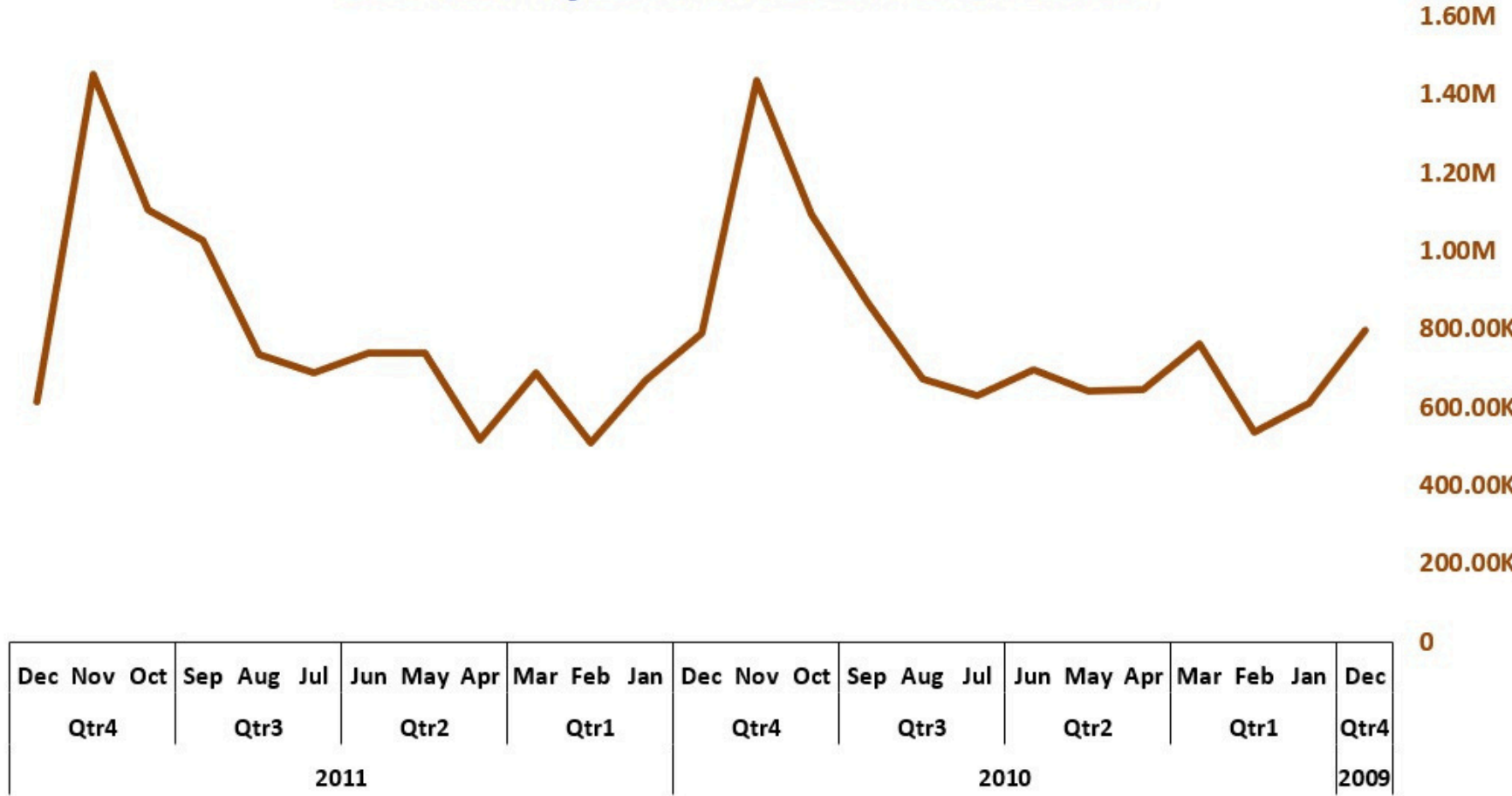
Launch targeted discounts and promotional campaigns in February to stimulate demand and improve sales.

How do monthly revenues trend across the two years?

Revenue & Quantity by Products Top 10



Revenue by Year & Month & Quarters



Which products generate the highest revenue and sales volume?

— Revenue $\neq$ Sales Volume

Insight:

Cake Stand generated the highest revenue despite having a lower sales volume than products such as Jumbo Bags.

Solution:

Treat customers purchasing high-revenue products as valuable customers and use targeted offers to encourage repeat purchases.

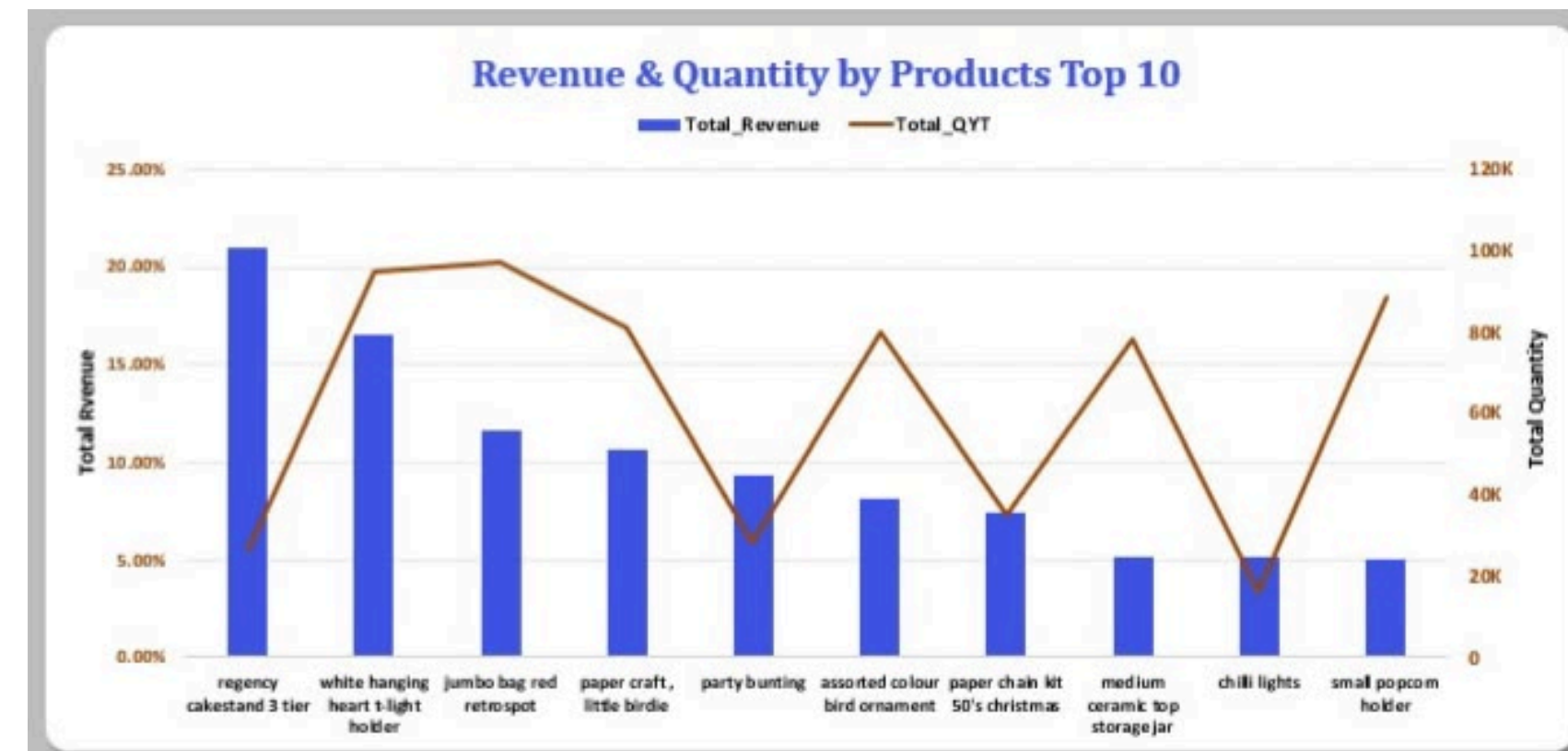
— High-Volume Products

Insight:

Jumbo Bag has the highest sales volume, indicating strong customer demand.

Solution:

Maintain sufficient stock levels to avoid stockouts and prevent lost sales.



How does revenue vary across days of the week?

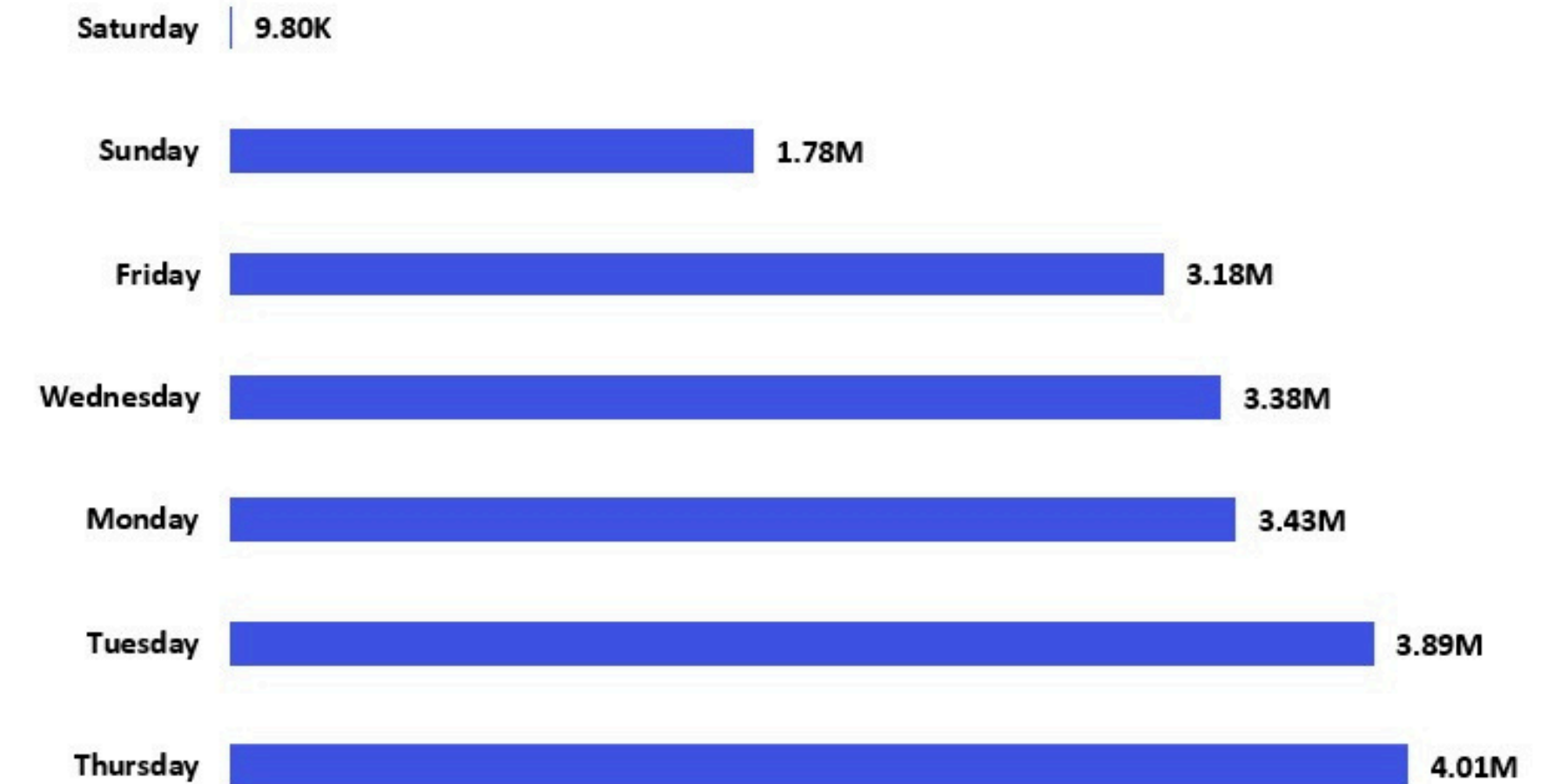
Insight

While most days generate millions in revenue, Saturday shows an exceptionally large drop compared with high-performing days like Tuesday and Thursday.

Solution

Run stronger promotions on Saturdays to stimulate demand, while ensuring sufficient stock levels on high-demand weekdays.

Sales Over The Days



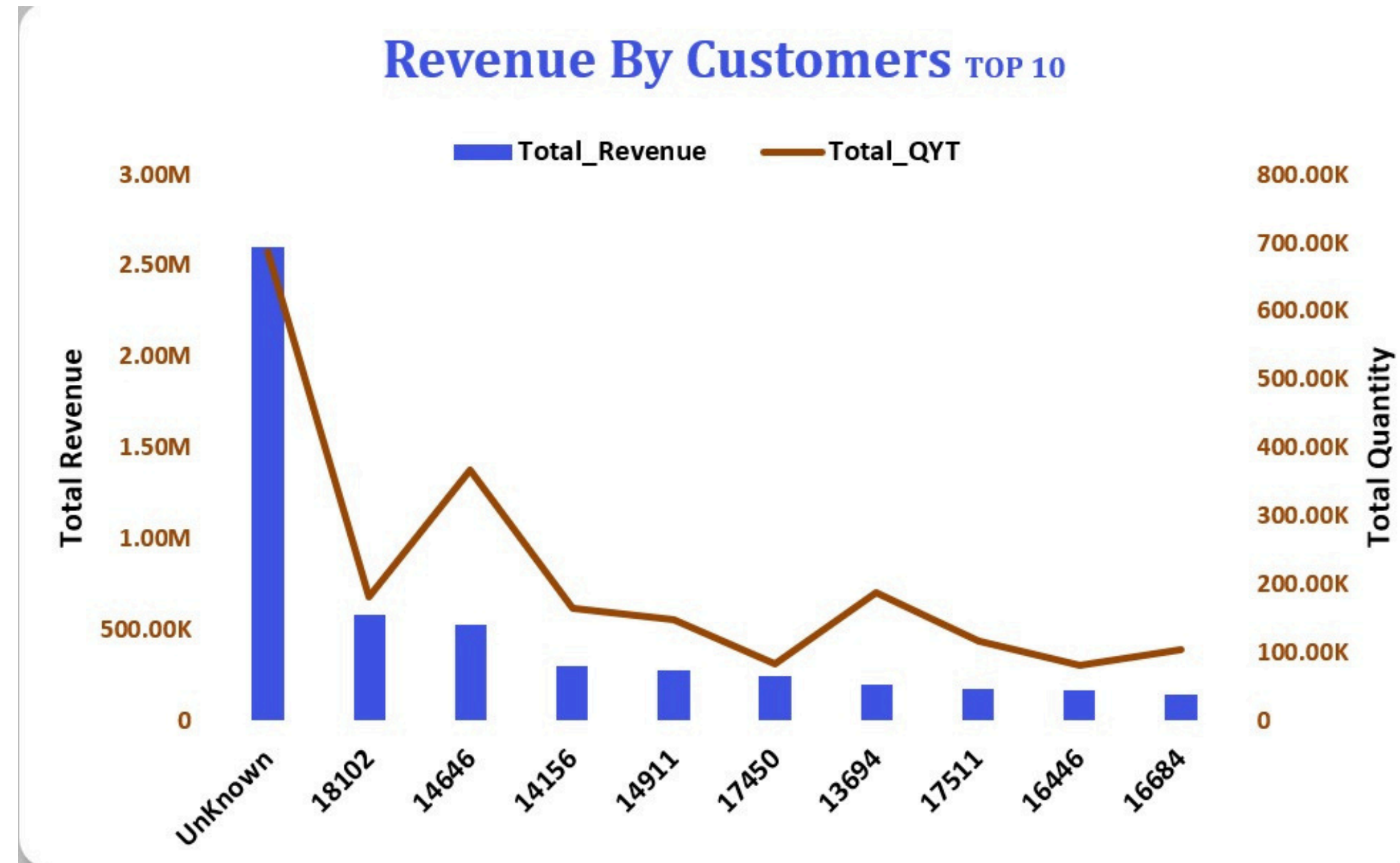
How much revenue is associated with missing Customer IDs?

Insight

A significant amount of revenue and sales volume comes from transactions with missing Customer IDs, making it difficult to identify and analyze these customers.

Solution

Improve customer data collection to better understand customer behavior and enable more effective targeting and retention strategies.



Who are the top customers, and how do their purchasing behaviors differ?

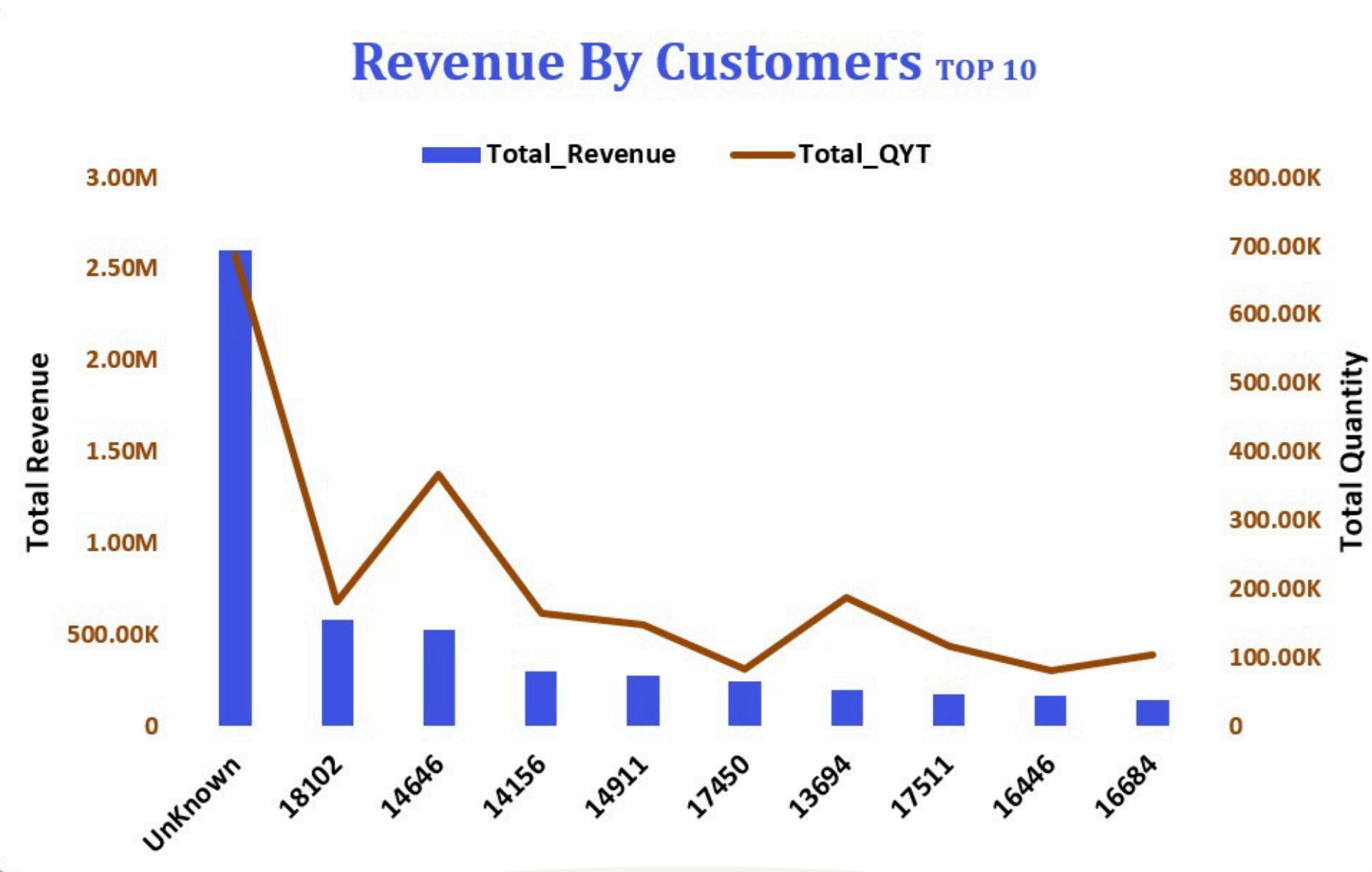
Insight

After excluding customers with missing IDs, Customer 18102 generated 580.99K in revenue with a lower quantity, while Customer 14646 generated 526.75K with a much higher quantity.

This shows that high-value customers can have different purchasing behaviors.

Solution

Offer quantity-based discounts to high-value customers who purchase fewer units, while maintaining loyalty benefits for high-volume customers to encourage repeat purchases and retention





Key insights

***Market: Revenue is highly concentrated in the UK.**

***Products: Revenue and sales volume are driven by different products.**

***Seasonality: Demand peaks around October–November and drops in February.**

***Daily Performance: Revenue varies significantly across weekdays.**

***Customers: High-value customers show different purchasing behaviors.**

***Data Quality: Missing Customer IDs limit customer-level analysis.**



Online Retail Performance



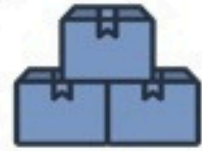
Date Update through:
12/9/2011



Total Revenue
19.68M



Total Orders
1.00M

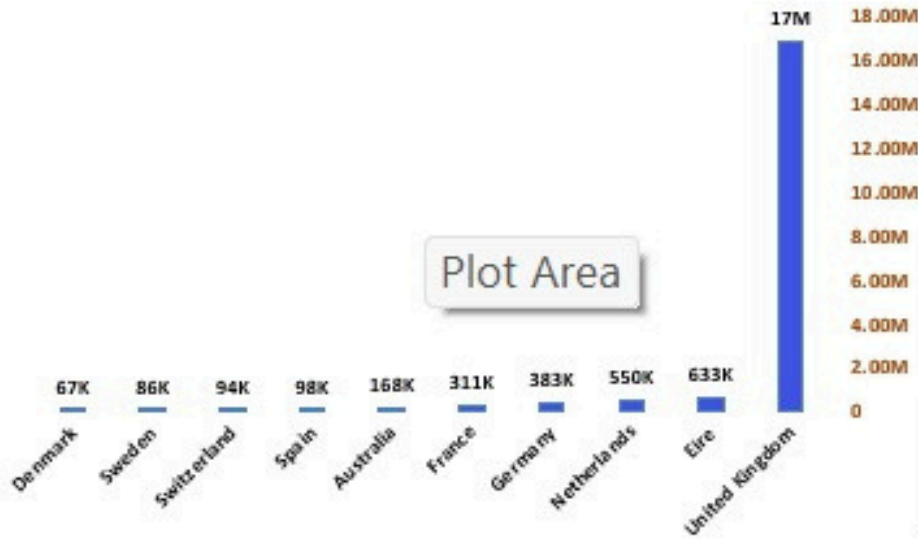


Total Quantity
11.18M

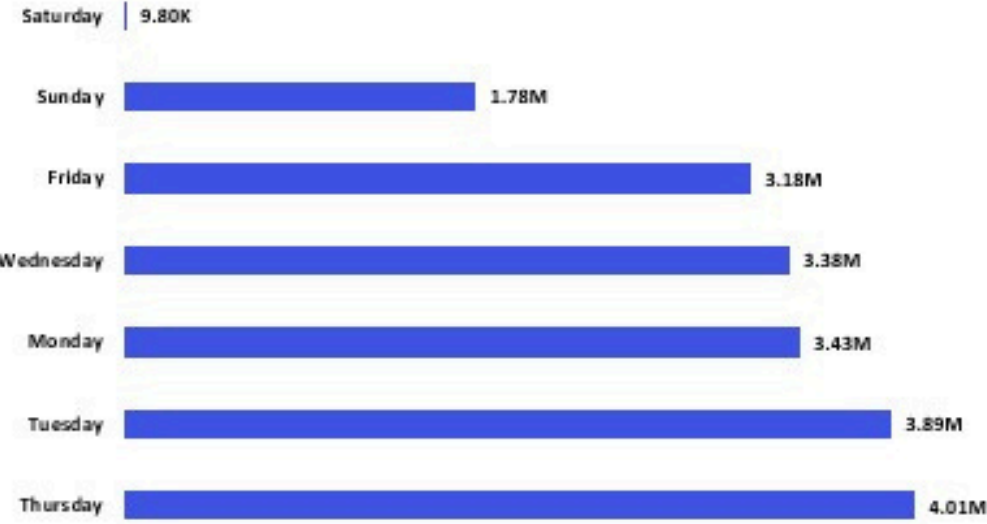


AVG Revenue
20

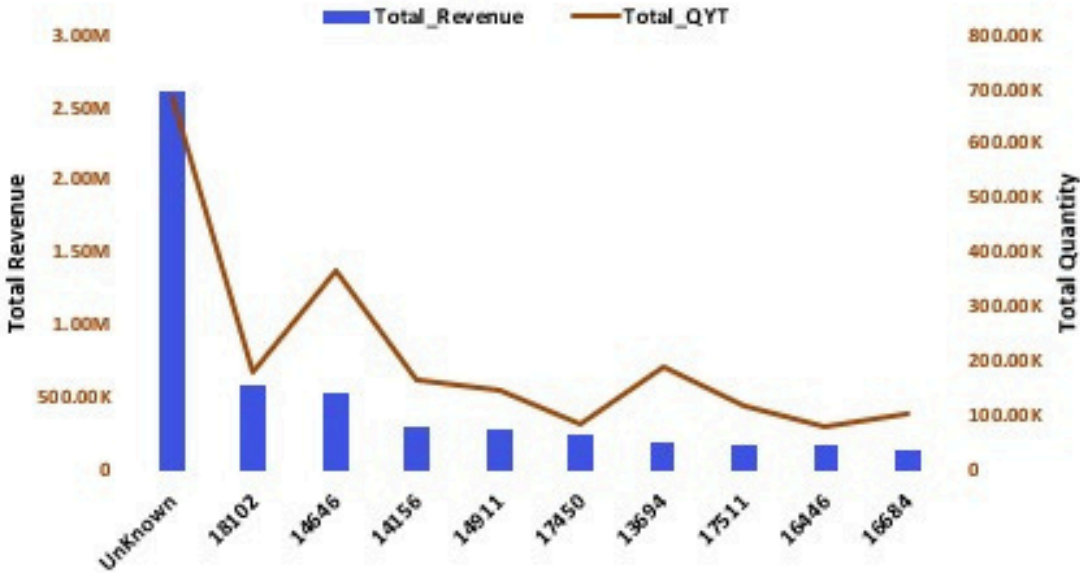
Revenue by Country



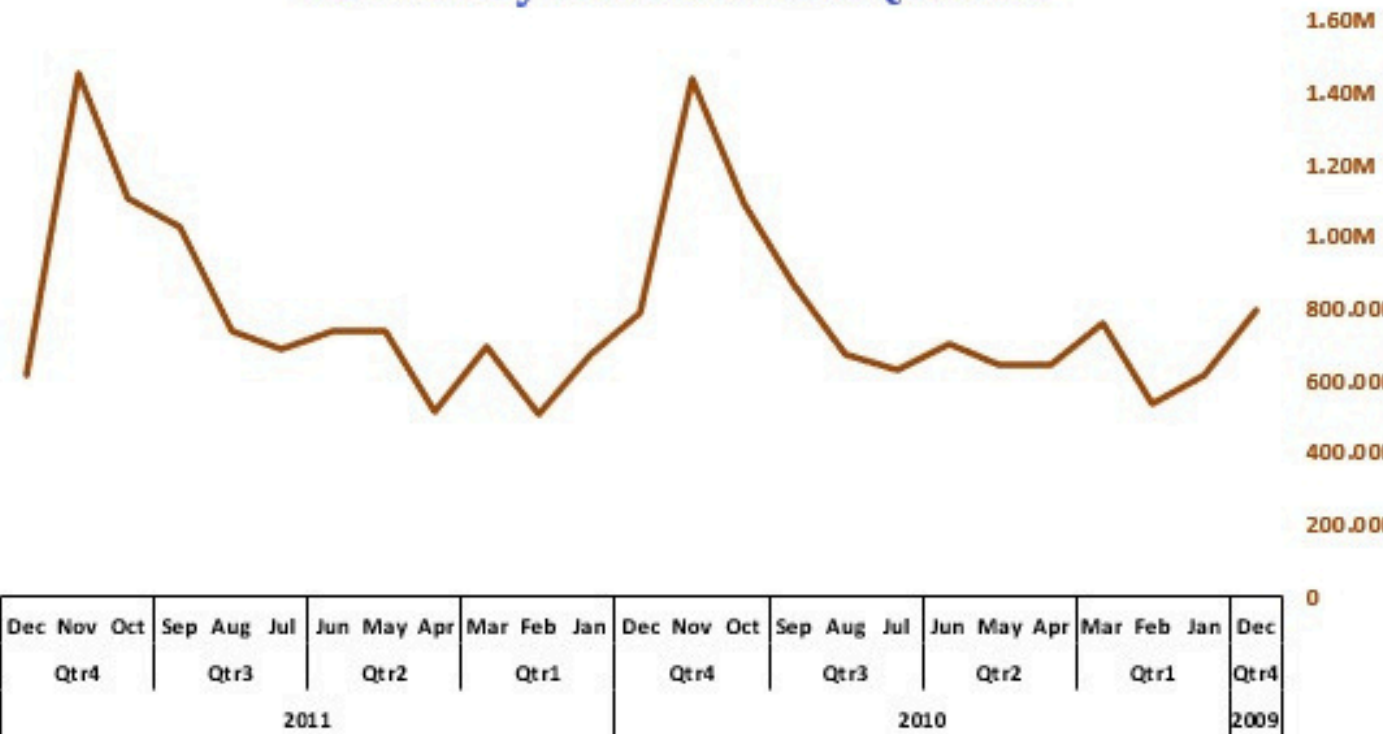
Sales Over The Days



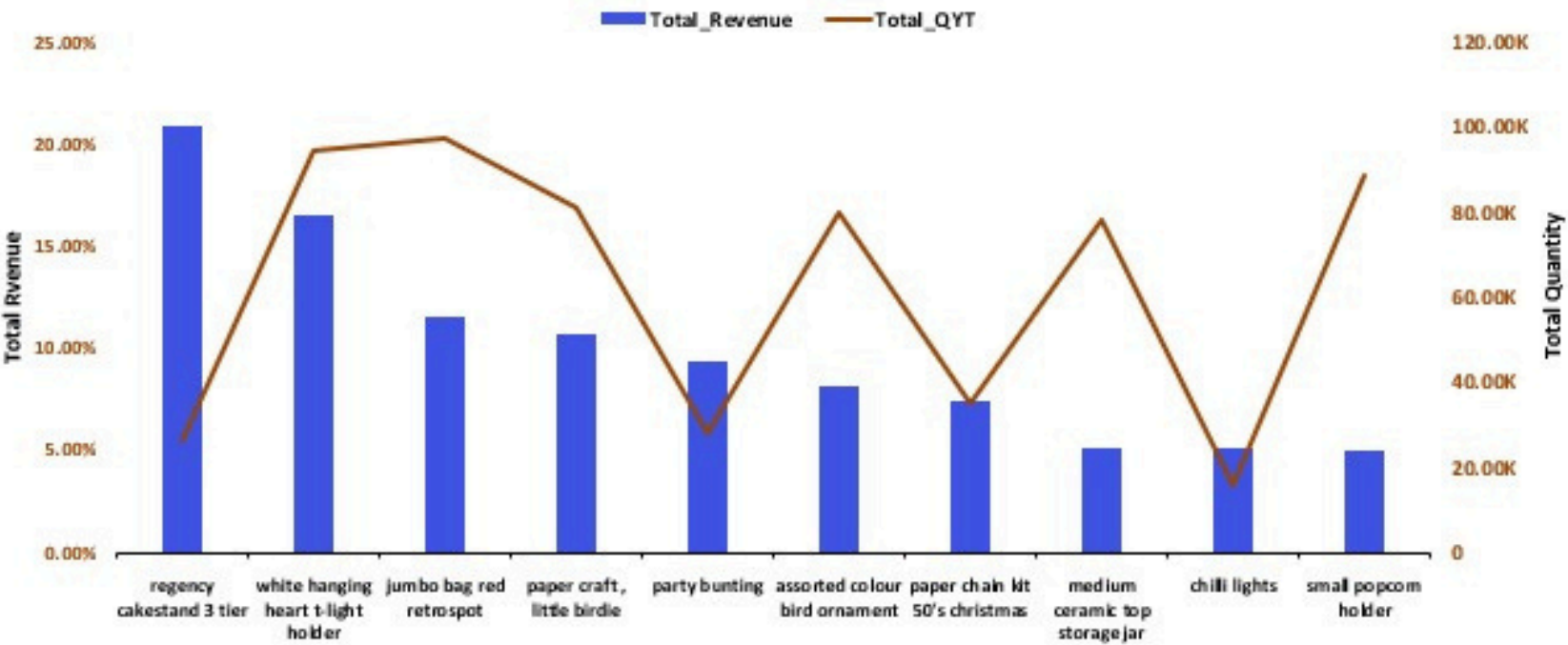
Revenue By Customers TOP 10



Revenue by Year & Month & Quarters



Revenue & Quantity by Products Top 10



Years

All Periods YEARS

2009 2010 2011

Quarters

All Periods QUARTERS

2009 2010

Q1 Q2 Q3 Q4 Q1 Q2 Q3

country

Canada

Channel Islands

Cyprus

Czech Republic

Denmark

Eire

European Community

Finland

France

Germany

Greece

Hong Kong

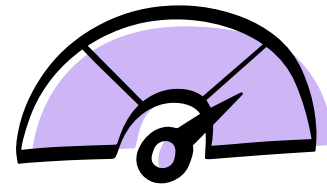
Iceland



Conclusion

Our analysis showed that revenue is strongly influenced by seasonality, product performance, and customer behavior. The UK is the main revenue-generating market, while some products and customers contribute significantly to sales.

These insights can help the company improve inventory planning, target marketing campaigns, and strengthen customer relationships.



KPIs



Total Revenue

19.68M



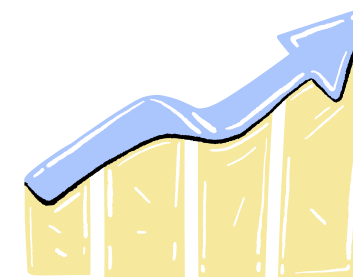
Total orders

1.00M



Total Quantity

11.18M



AVG Revenue

20M

Thank
you

